

Lab # 01

Installation of the Ubuntu Live Server

Name: Sara Ajmal

Registration Number: FA24B1-SE-30

Course Code: BSE-665

Section: B

GitHub Username: saraajmal30

Date: 19st, Sep, 2026

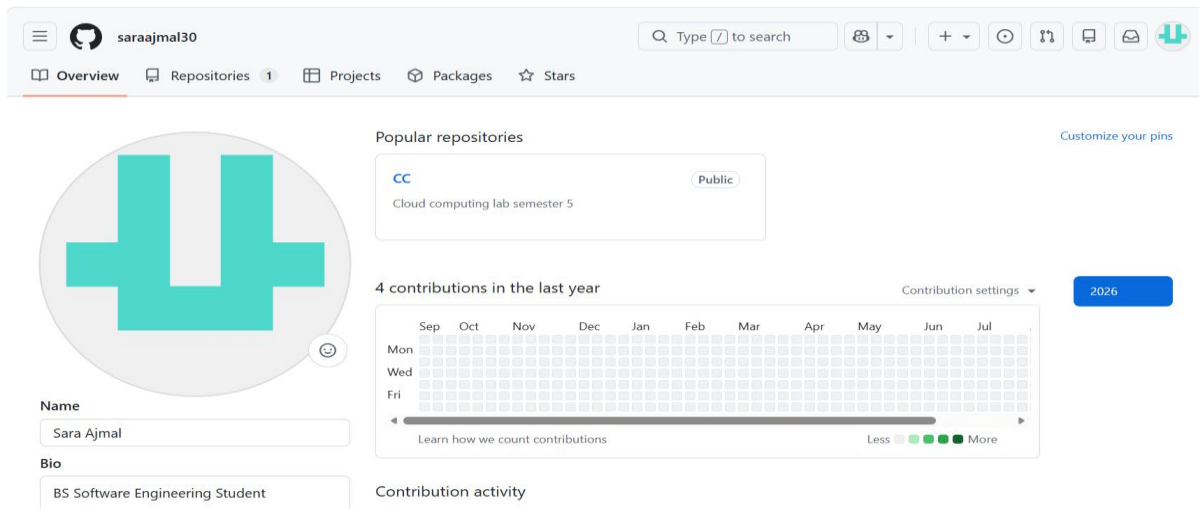
Submitted to: Sir Shoaib

Getting Started

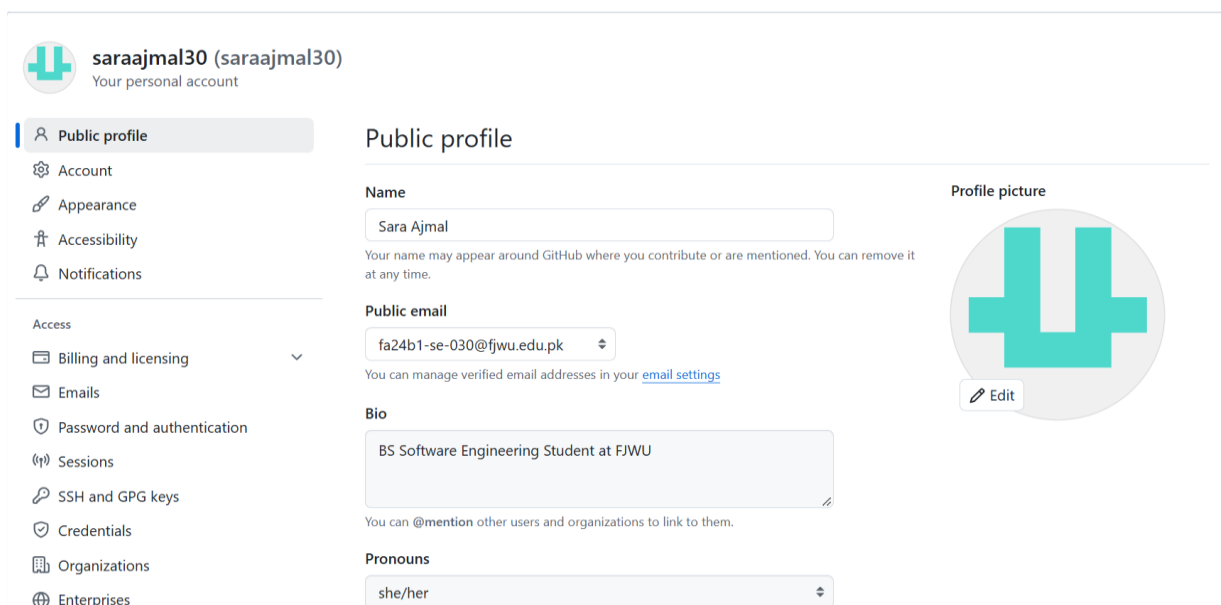
 saraajmal30	Create README.md	550a4ea
	Labs/Lab01	Create README.md
	README.md	Create README.md

- Demonstrates the required Labs/Lab01/ directory structure created inside the repository.

Task 1: Create and Configure Your GitHub Account

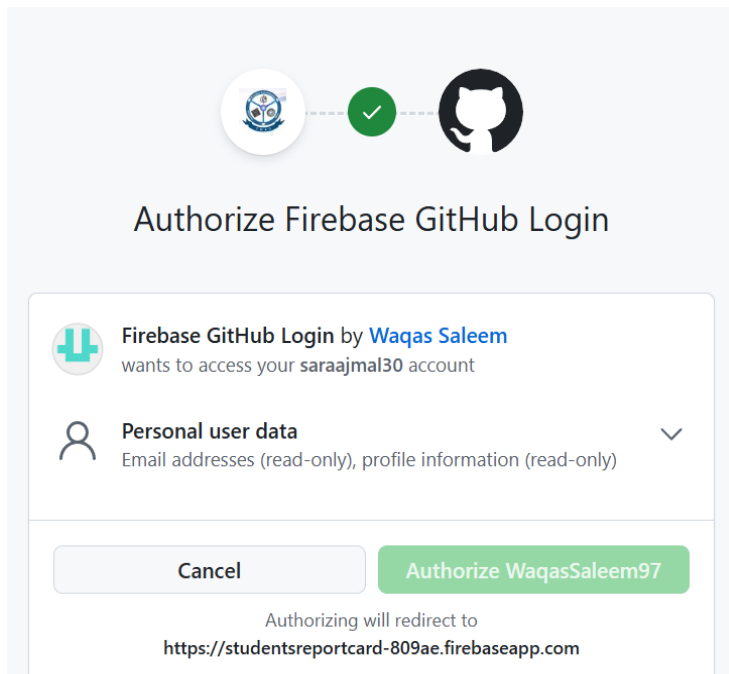


- Demonstrates the created public GitHub profile displaying the username saraajmal30



- Demonstrates the completed Name field under GitHub Public Profile settings displaying the student's full actual name.

Task 2: Register in the Student Marks Portal



- Demonstrates the Student Marks Portal login page prior to authenticating via GitHub.



Complete registration

Enter your registration number, course and section.
The enrollment requires approval.

Registration number

2024-BSE-30

Allowed: 2024-BSE-01 to 2024-BSE-99.

Course

SE-665 — Cloud Computing

Section

Section B

Submit enrollment

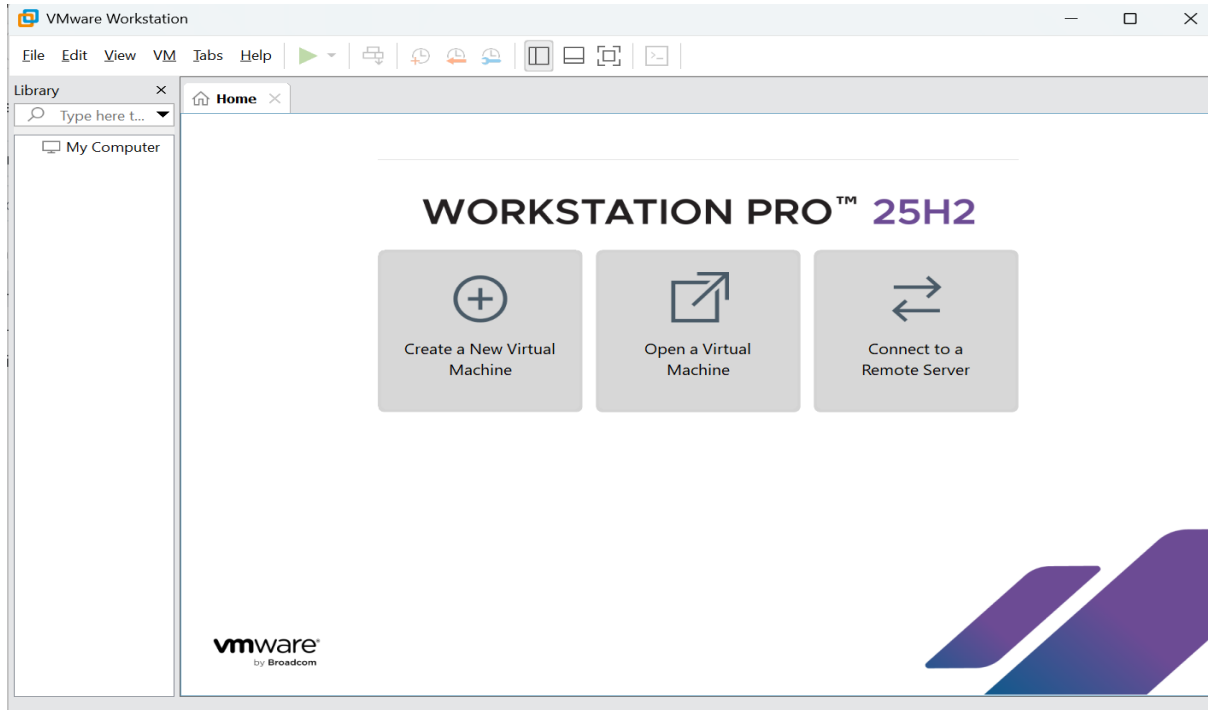
Sign out

- Demonstrates student GitHub profile information and registration details integrated inside the portal.

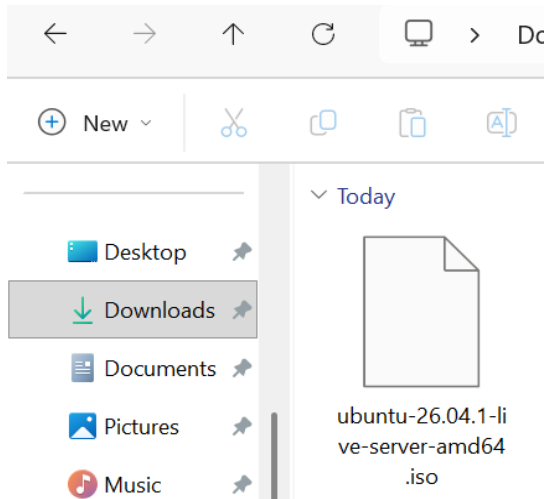


- Demonstrates successful submission of course enrollment awaiting approval inside the Student Marks Portal.

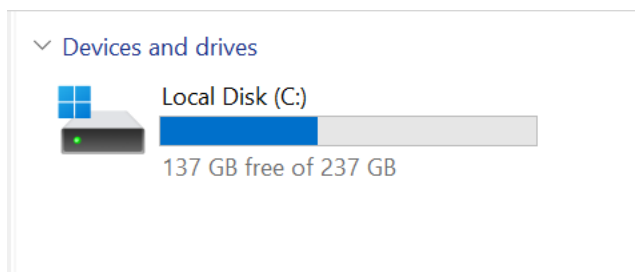
Task 3: Download VMware Workstation Pro and Ubuntu Server



- Demonstrates VMware Workstation Pro installed and running on the Windows host machine.

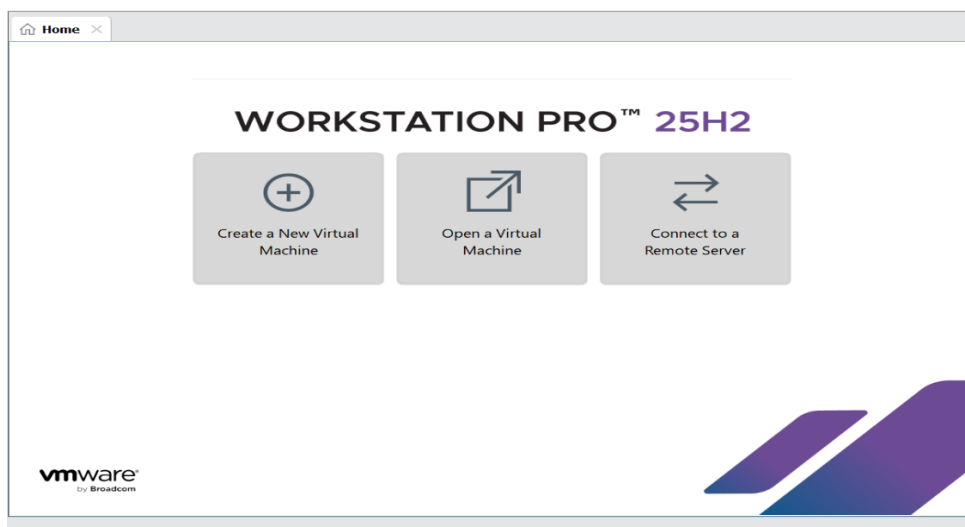


- Displays the downloaded Ubuntu Server LTS ISO file inside the host system file manager.



- Demonstrates sufficient available local disk storage (minimum 20 GB) for virtual machine creation.

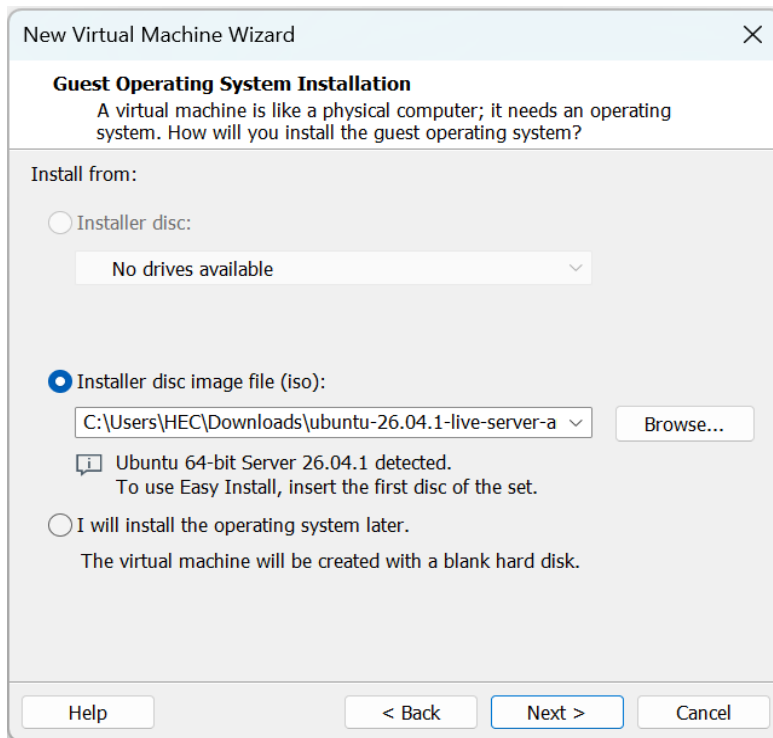
Task 4: Create the Ubuntu Server Virtual Machine



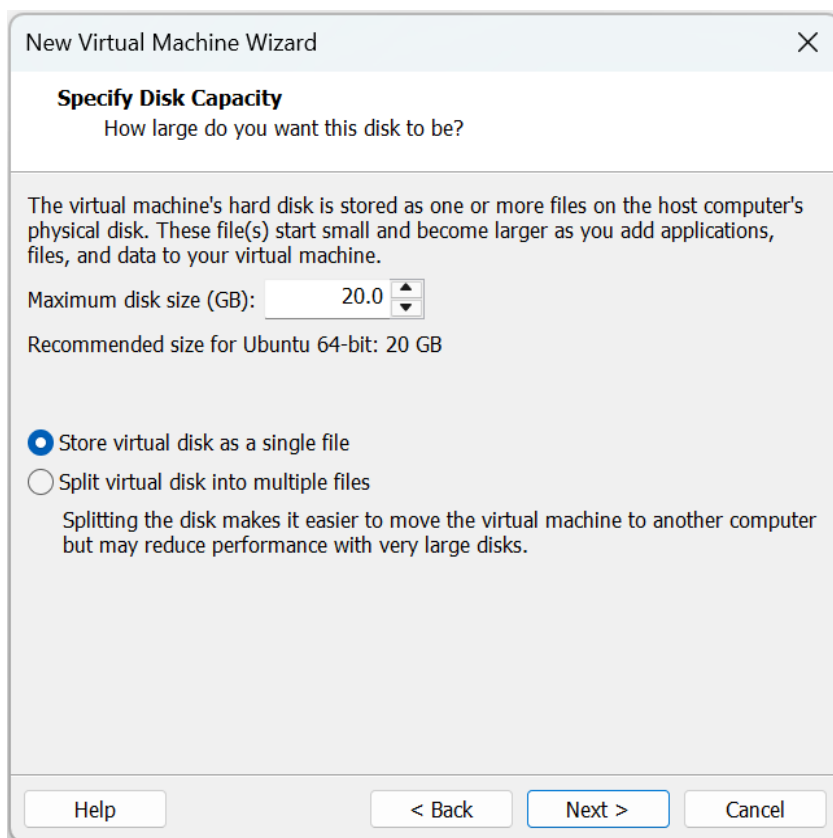
- Demonstrates opening the VMware New Virtual Machine Wizard to begin virtual machine setup.



- Demonstrates selecting the Typical configuration option for the virtual machine setup.



- Displays the selected local file path for the Ubuntu Server ISO installer image.

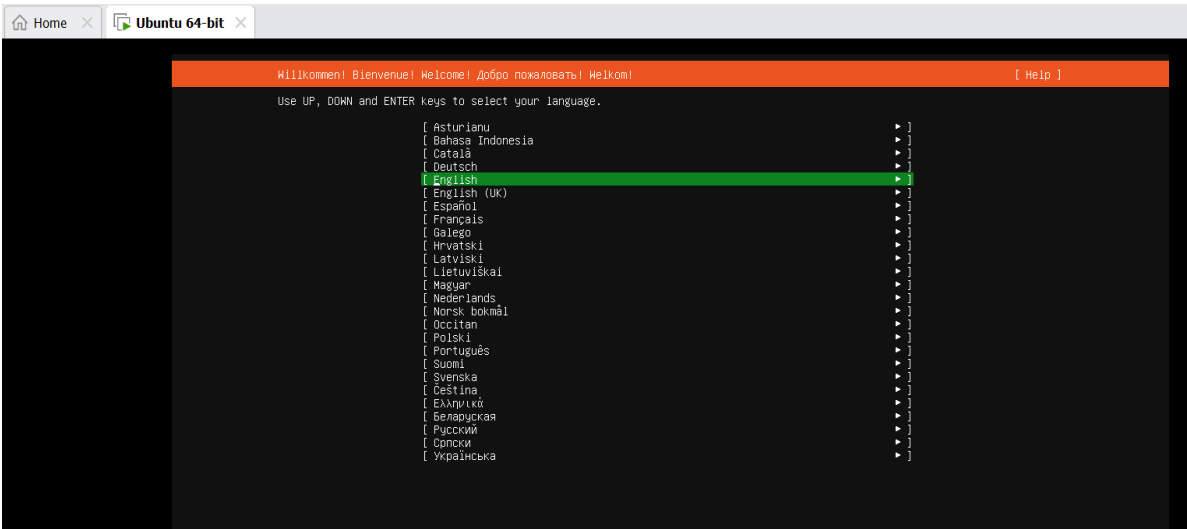


- Demonstrates the final virtual machine configuration summary prior to creation.

```
.
Begin: Configuring fstab... .. done.
Begin: Setting up locales... .. done.
Begin: Setting up automatic login... .. done.
Begin: Setting hostname... .. done.
Begin: Setting up console keyboard... .. done.
Begin: Applying desktop settings... .. done.
Begin: Regenerating SSL certificate... .. done.
Begin: Loading preseed file... .. done.
Begin: Adding live session user... .. ^[Spawwd: password changed.
done.
Begin: Setting up init... .. done.
Begin: Configuring accessibility options... .. done.
Begin: Disabling update-notifier... .. done.
Begin: Enabling detection of crashes... .. done.
Begin: Disabling unnecessary KDE services... .. done.
Begin: Disabling the lock screen for LXQt (LP: #2000787)... .. done.
Begin: Generating APT cache for cdrom .. Ign:1 file:/cdrom resolute InRelease
Get:2 file:/cdrom resolute Release [664 B]
Get:2 file:/cdrom resolute Release [664 B]
Get:3 file:/cdrom resolute Release.gpg [228 B]
Get:3 file:/cdrom resolute Release.gpg [228 B]
Get:4 file:/cdrom resolute/main amd64 Packages [39.1 kB]
Get:5 file:/cdrom resolute/restricted amd64 Packages [13.9 kB]
^[
```

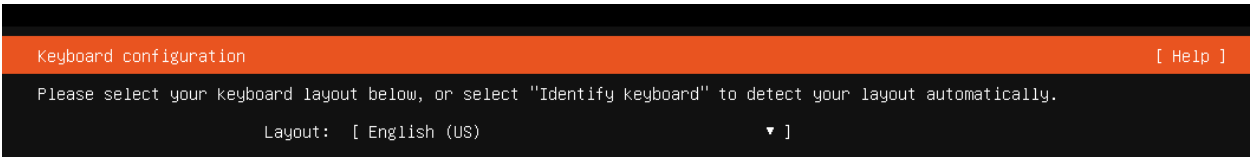
- Demonstrates the Ubuntu Server installer boot screen upon powering on the virtual machine.

Task 5: Choose the Installation Language

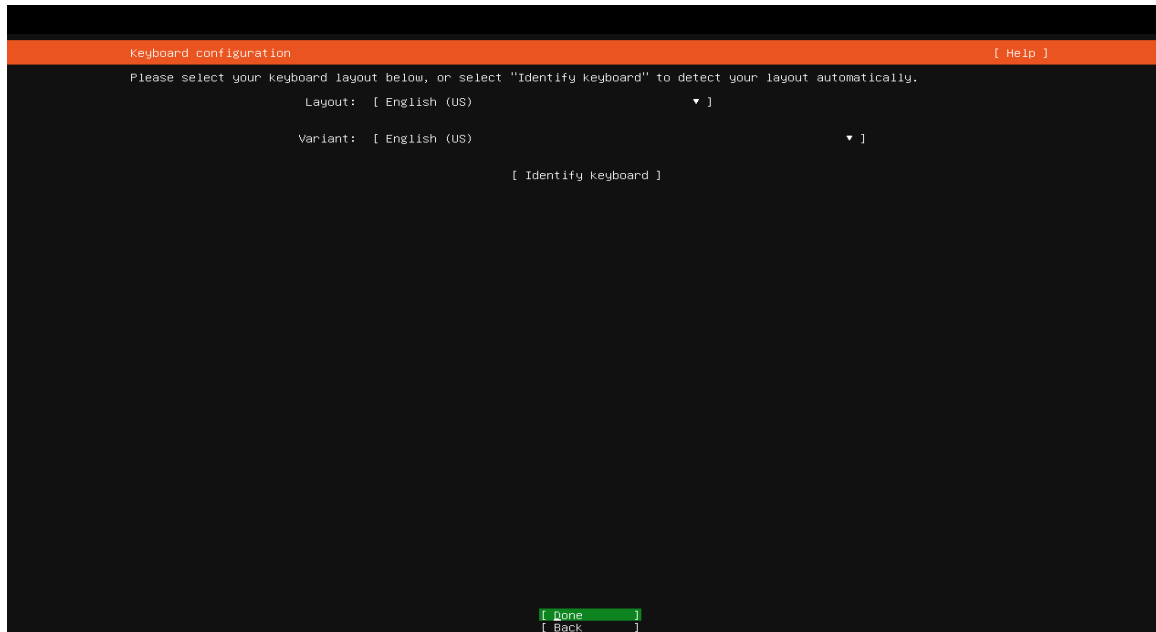


- Demonstrates selecting English as the primary installer language.

Task 6: Configure the Keyboard Layout

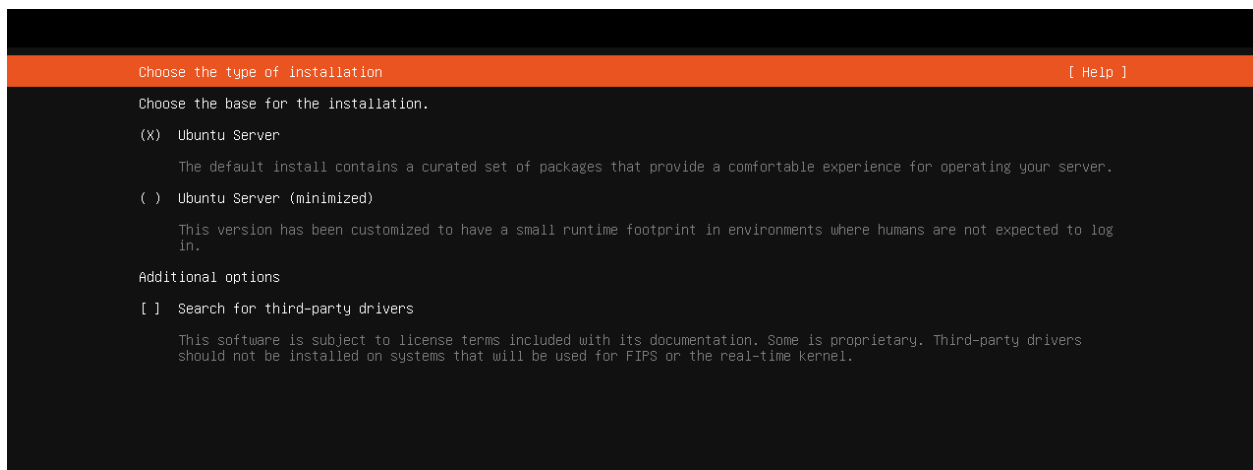


- Demonstrates selecting the primary keyboard layout (English US) during the Ubuntu Server installation setup.



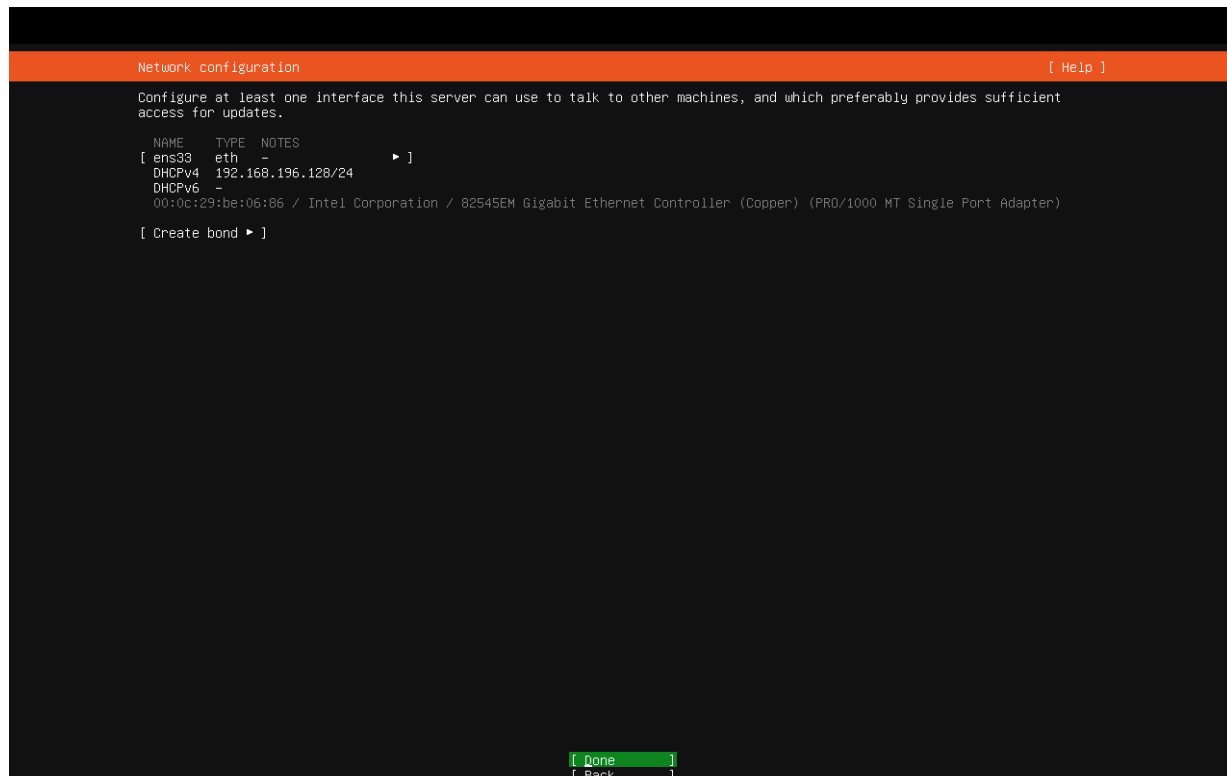
- Displays selecting the matching keyboard layout variant (English US) before proceeding with system setup.

Task 7: Choose the Ubuntu Server Installation Type

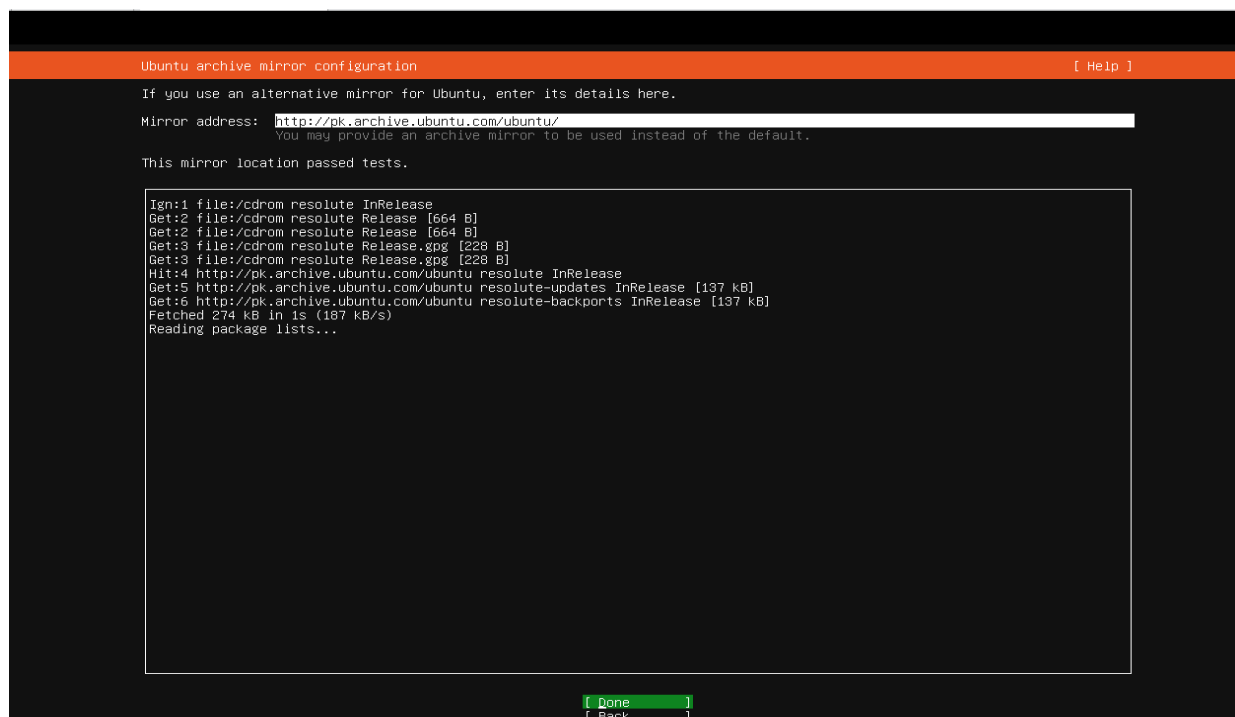


- Demonstrates selecting the standard base Ubuntu Server installation option instead of MAAS deployment.

Task 8: Configure the Network

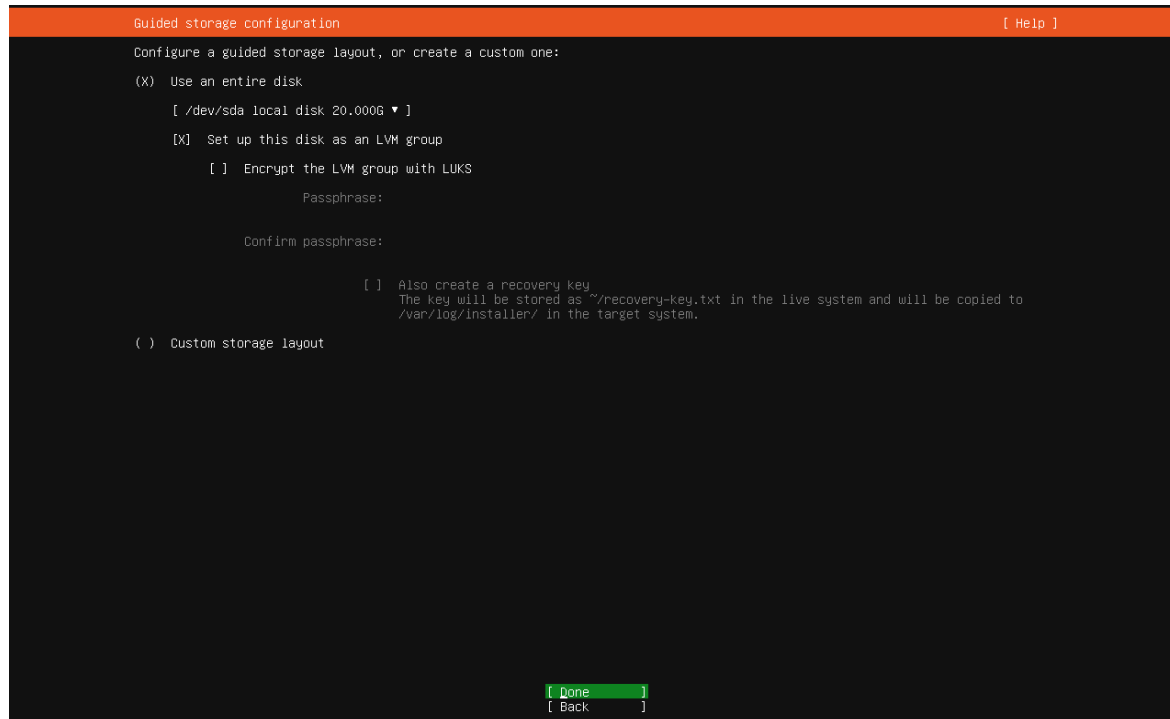


- Demonstrates detection of the primary Ethernet network interface (ens33) by the Ubuntu Server installer.

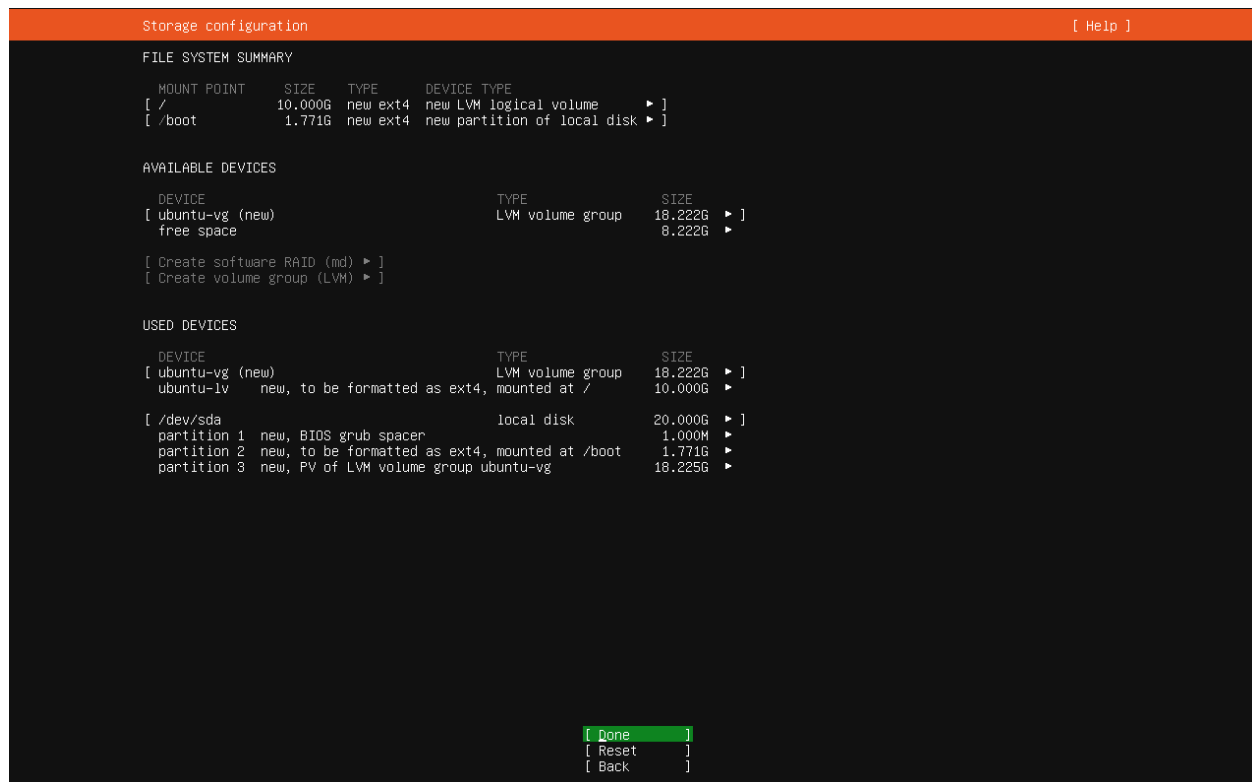


- Displays the active IPv4 address automatically assigned to the network interface via DHCP.

Task 9: Configure Storage

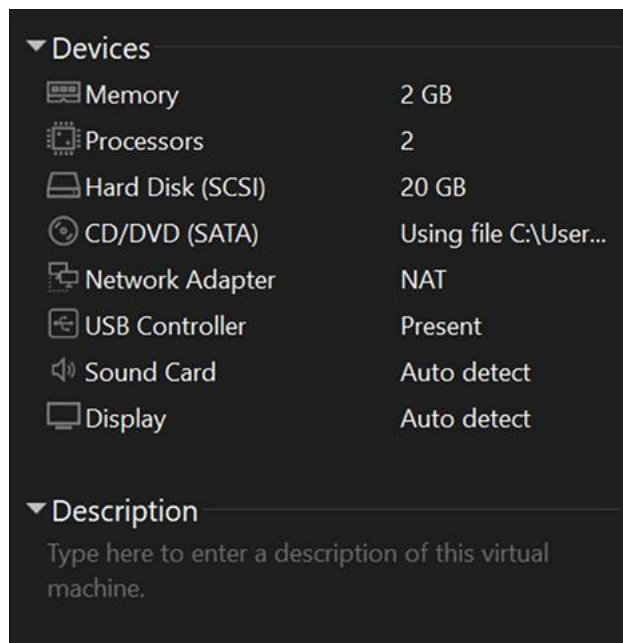


- Demonstrates selecting the guided storage option to use the allocated VMware virtual disk for installation.

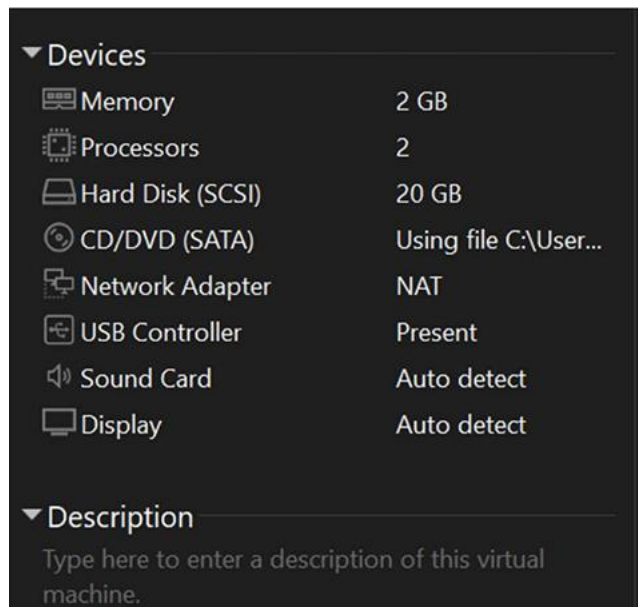


- Displays the reviewed filesystem storage layout and partition summary prior to applying changes.

Task 10: Select the Installation Disk

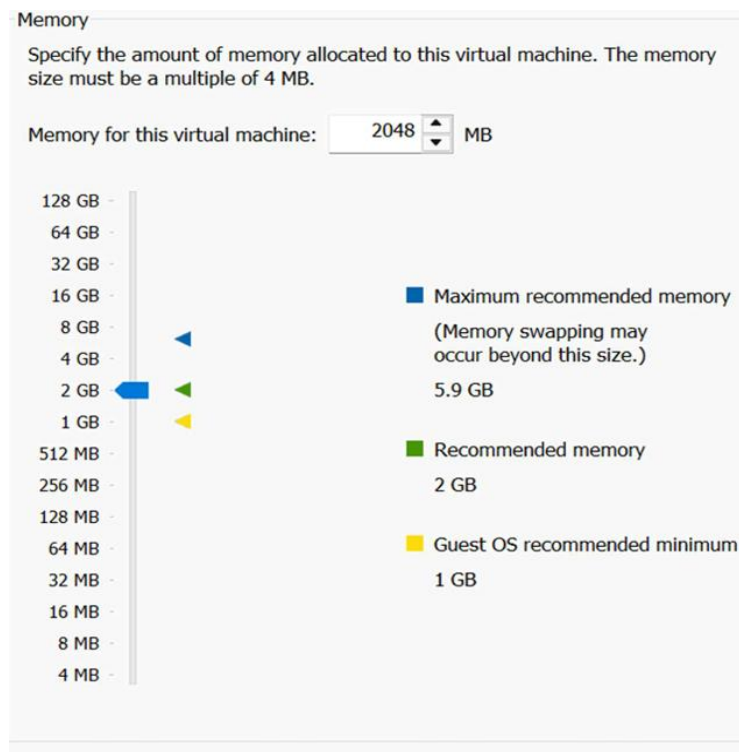


- Demonstrates selecting the primary target virtual disk for the Ubuntu Server installation.



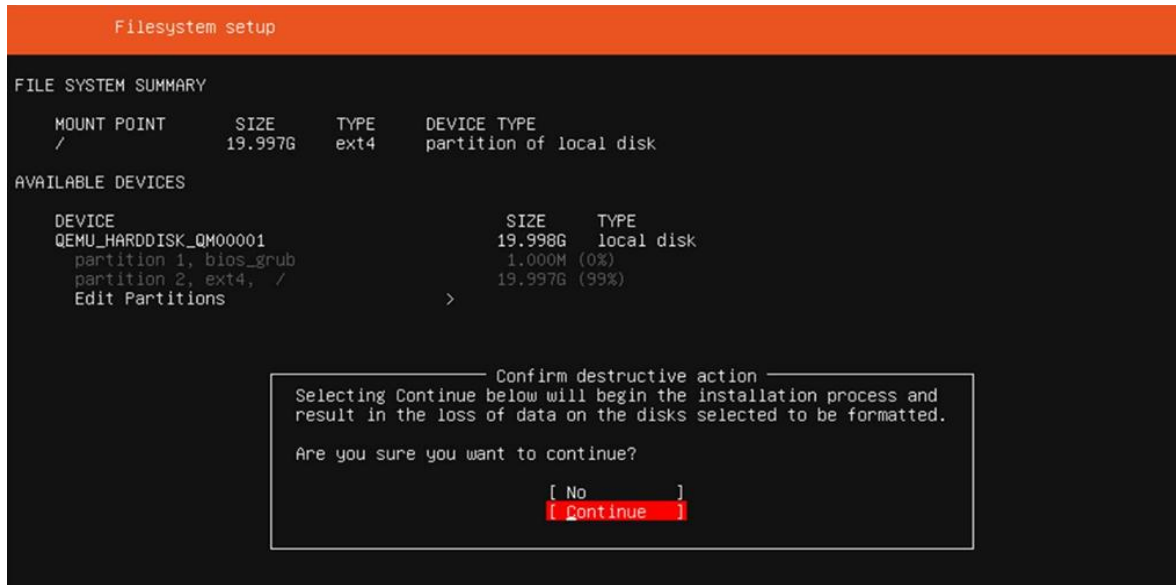
- Displays the target virtual disk name and allocated capacity to verify installation on the correct VMware storage.

Task 11: Review the Partitions

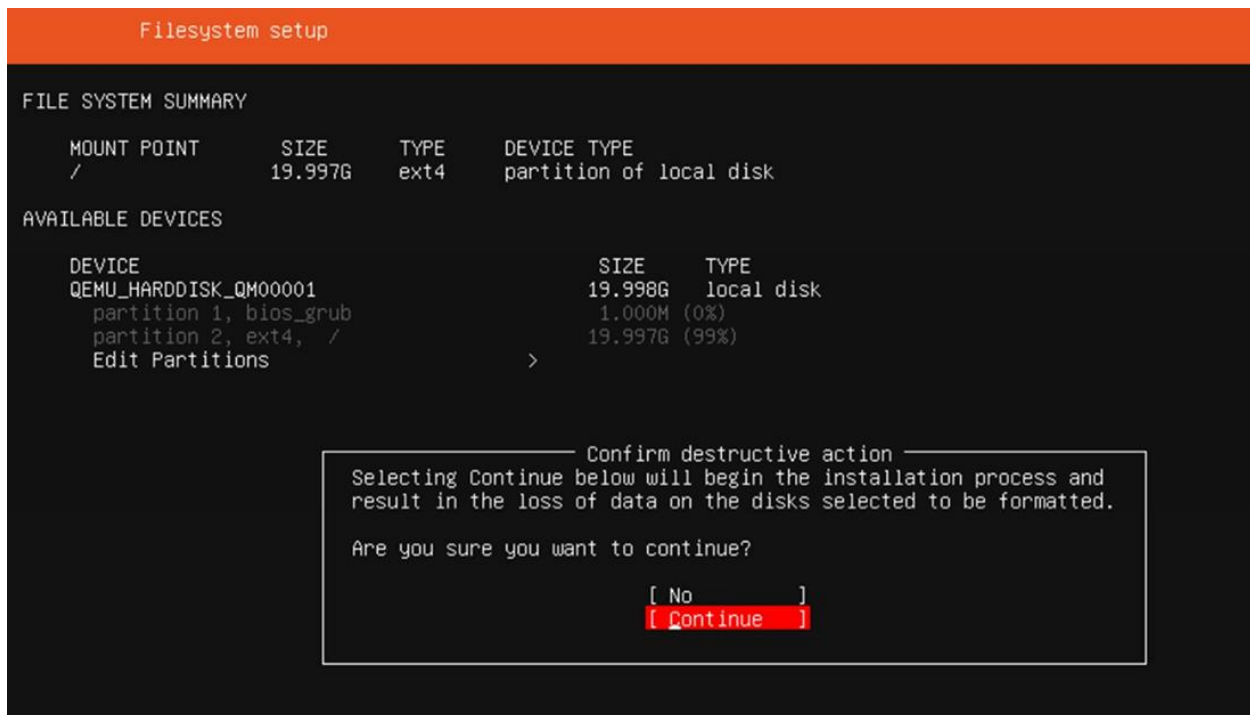


- Demonstrates the proposed partition layout and filesystem summary on the target VMware virtual disk.

Task 12: Confirm the Storage Changes



- Displays the installer's destructive-action warning prompt prior to overwriting the virtual disk.



- Demonstrates confirming the storage action to proceed with disk partitioning and formatting.

Task 13: Configure the Ubuntu Profile

Profile configuration

[Help]

Enter the username and password you will use to log in to the system. You can configure SSH access on a later screen, but a password is still needed for sudo.

Your name:

Your servers name:

The name it uses when it talks to other computers.

Pick a username:

Choose a password:

Confirm your password:

[Done]

- Demonstrates entering the student's full actual name in the profile configuration screen.

Profile configuration

[Help]

Enter the username and password you will use to log in to the system. You can configure SSH access on a later screen, but a password is still needed for sudo.

Your name:

Your servers name:

The name it uses when it talks to other computers.

Pick a username:

Username missing

Choose a password:

Confirm your password:

[Done]

- Demonstrates setting the server hostname to ubuntu during system configuration.

Profile configuration

[Help]

Enter the username and password you will use to log in to the system. You can configure SSH access on a later screen, but a password is still needed for sudo.

Your name:

Your servers name:
The name it uses when it talks to other computers.

Pick a username:

Choose a password:

Confirm your password:

[Done]

- Demonstrates configuring the primary system username to match the student's GitHub handle (saraajmal30).

Profile configuration

[Help]

Enter the username and password you will use to log in to the system. You can configure SSH access on a later screen, but a password is still needed for sudo.

Your name:

Your servers name:
The name it uses when it talks to other computers.

Pick a username:

Choose a password:

Confirm your password:

[Done]

Demonstrates the fully completed profile setup page featuring hostname and user credentials.

Task 14: Complete the Software Installation

Installing system

[Help]

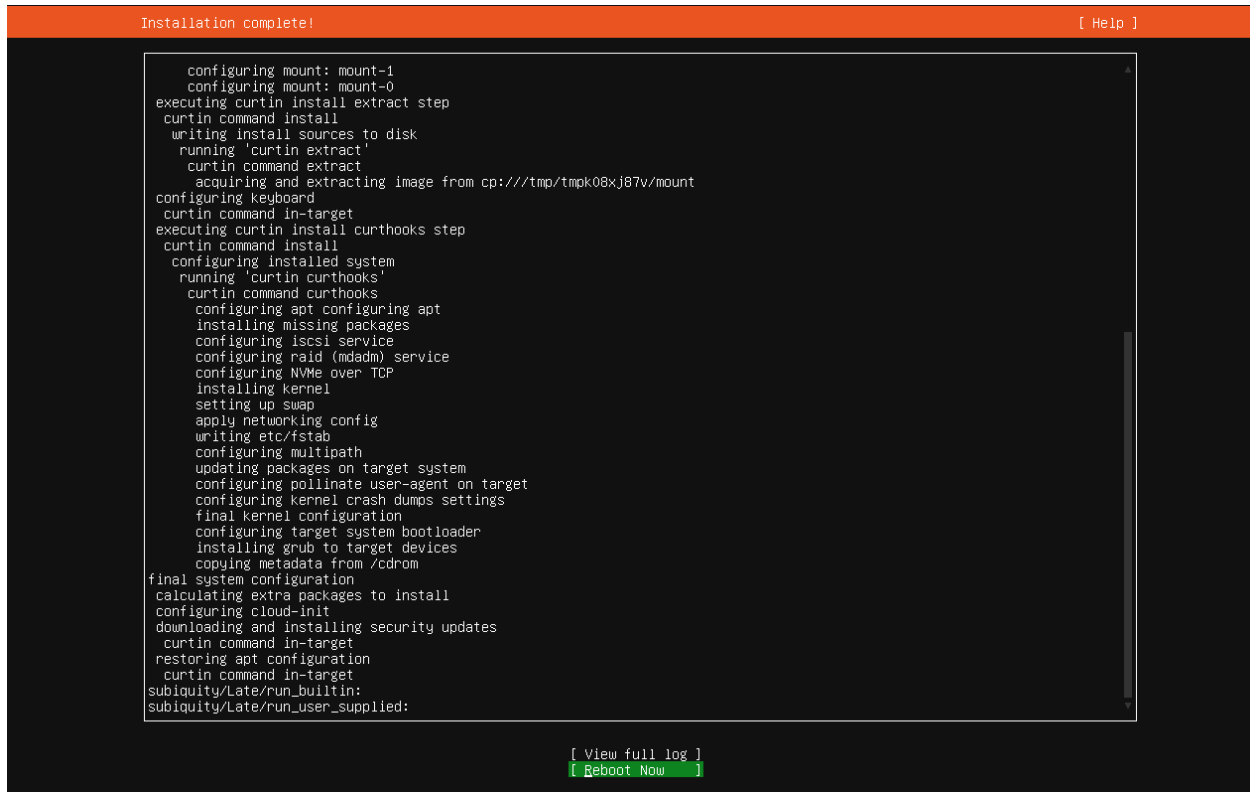
```

subiquity/2dev/apply_autoinstall_config:
subiquity/Ad/apply_autoinstall_config:
subiquity/late/apply_autoinstall_config:
configuring apt
curtin command in-target
installing system
executing curtin install initial step
executing curtin install partitioning step
curtin command install
configuring storage
running 'curtin block-meta simple'
curtin command block-meta
removing previous storage devices
configuring disk: disk-sda
configuring partition: partition-0
configuring partition: partition-1
configuring partition: partition-2
configuring format: format-0
configuring lvm_volgroup: lvm_volgroup-0
configuring lvm_partition: lvm_partition-0
configuring format: format-1
configuring mount: mount-1
configuring mount: mount-0
executing curtin install extract step
curtin command install
writing install sources to disk
running 'curtin extract'
curtin command extract
acquiring and extracting image from cp:///tmp/tmpk08xj87v/mount
configuring keyboard
curtin command in-target
executing curtin install curthooks step
curtin command install
configuring installed system
running 'curtin curthooks'
curtin command curthooks
configuring apt configuring apt
installing missing packages
configuring iscsi service
configuring raid (mdadm) service
configuring NVMe over TCP
installing kernel |
  
```

[View full log]

- Displays the ongoing system installation progress and package extraction on the virtual machine.

Task 15: Reboot and Verify the Ubuntu Identity

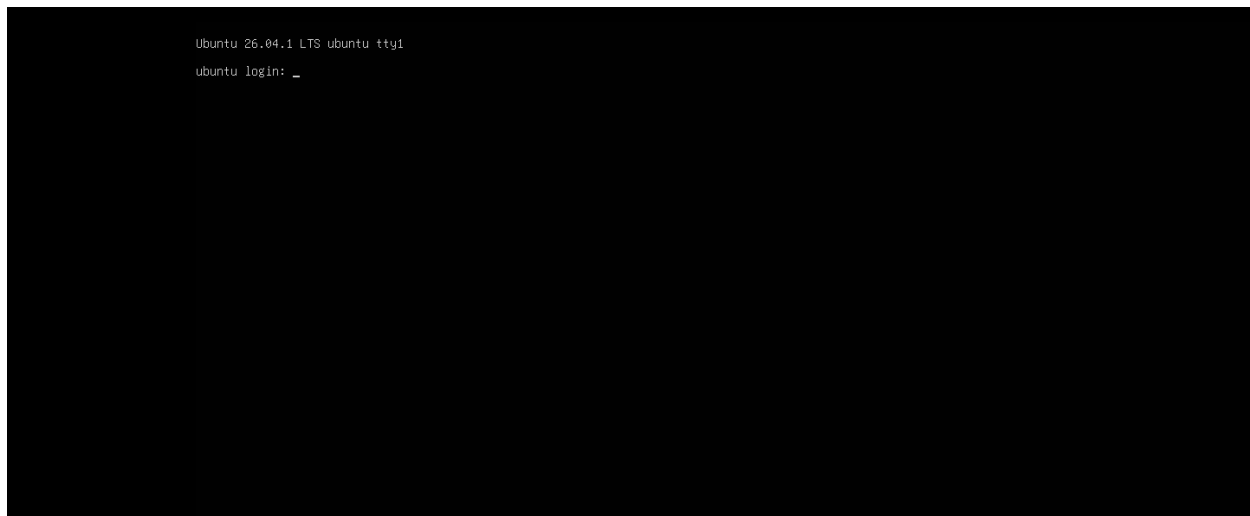


```
Installation complete! [ Help ]

configuring mount: mount-1
configuring mount: mount-0
executing curtin install extract step
curtin command install
writing install sources to disk
running 'curtin extract'
curtin command extract
acquiring and extracting image from cp:///tmp/tmpk08xj87v/mount
configuring keyboard
curtin command in-target
executing curtin install curthooks step
curtin command install
configuring installed system
running 'curtin curthooks'
curtin command curthooks
configuring apt configuring apt
installing missing packages
configuring iscsi service
configuring raid (mdadm) service
configuring NVMe over TCP
installing kernel
setting up swap
apply networking config
writing etc/fstab
configuring multipath
updating packages on target system
configuring pollinate user-agent on target
configuring kernel crash dumps settings
final kernel configuration
configuring target system bootloader
installing grub to target devices
copying metadata from /cdrom
final system configuration
calculating extra packages to install
configuring cloud-init
downloading and installing security updates
curtin command in-target
restoring apt configuration
curtin command in-target
subiquity/Late/run_builtin:
subiquity/Late/run_user_supplied:
```

[View full log]
[Reboot Now]

- Demonstrates the installation complete notification prior to system reboot.



```
Ubuntu 26.04.1 LTS ubuntu tty1
ubuntu login: _
```

- Displays the fresh Ubuntu Server TTY login prompt after booting up.

```

Ubuntu 26.04.1 LTS ubuntu tty1

ubuntu login: saraajmal30
Password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-31-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Sun Sep 20 07:58:28 PM UTC 2026

System load:  0.02          Processes:      248
Usage of /:   50.6% of 9.75GB Users logged in:   0
Memory usage: 21%          IPv4 address for ens33: 192.168.196.129
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

13 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

saraajmal30@ubuntu:~$

```

- Demonstrates successful local terminal login using user credentials.

```
saraajmal30@ubuntu:~$ _
```

- Verifies system identity displaying terminal prompt saraajmal30@ubuntu:~\$.

Task 16: Find the Ubuntu Server IP Address

```

saraajmal30@ubuntu:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 00:0c:29:be:06:86 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    altname enx00c29be0686
    inet 192.168.196.129/24 metric 100 brd 192.168.196.255 scope global dynamic ens33
        valid_lft 1270sec preferred_lft 1270sec
    inet6 fe80::20c:29ff:febe:686/64 scope link proto kernel_l1
        valid_lft forever preferred_lft forever
saraajmal30@ubuntu:~$

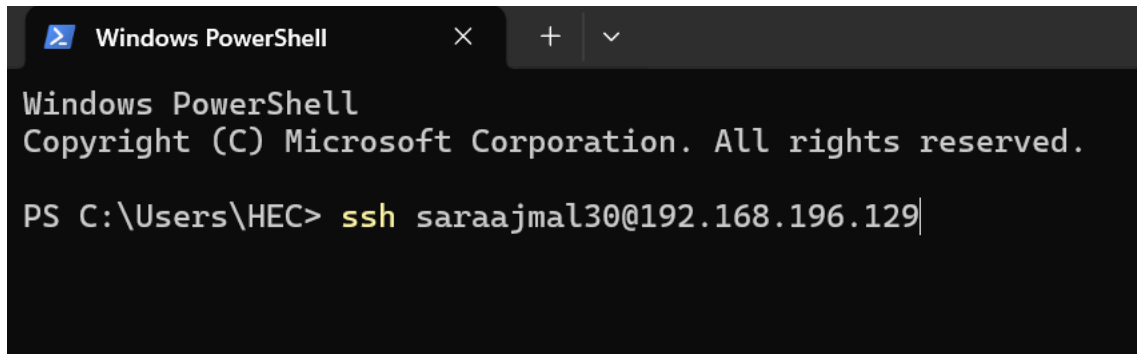
```

- Demonstrates executing the ip addr command inside the Ubuntu Server terminal.

```
saraajmal30@ubuntu:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 00:0c:29:be:06:86 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    altname enx00c29be0686
    inet 192.168.196.129/24 metric 100 brd 192.168.196.255 scope global dynamic ens33
        valid_lft forever preferred_lft forever
```

- Displays the assigned network interface IPv4 address alongside the saraajmal30@ubuntu:~\$ identity prompt.

Task 17: Connect to Ubuntu Server from Windows



```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\Users\HEC> ssh saraajmal30@192.168.196.129
```

- Demonstrates initiating an SSH remote connection from Windows terminal to the Ubuntu Server.

```
PS C:\Users\HEC> ssh saraajmal30@192.168.1.9
The authenticity of host '192.168.1.9 (192.168.1.9)' can't be established.
ED25519 key fingerprint is SHA256:eBUSEb1Bt92JR7tjRE5L/x8gUjTSorCh1VqUAVqmxgw.
This host key is known by the following other names/addresses:
    C:\Users\HEC/.ssh/known_hosts:1: 192.168.196.129
Are you sure you want to continue connecting (yes/no/[fingerprint])? |
```

- Displays accepting the host key fingerprint to establish a trusted SSH connection.

```

PS C:\Users\HEC> ssh saraajmal30@192.168.1.9
The authenticity of host '192.168.1.9 (192.168.1.9)' can't be established.
ED25519 key fingerprint is SHA256:eBUSEb1Bt92JR7tjRE5L/x8gUjTSoRch1VqUAVqmxgw.
This host key is known by the following other names/addresses:
  C:\Users\HEC/.ssh/known_hosts:1: 192.168.196.129
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.1.9' (ED25519) to the list of known hosts.
saraajmal30@192.168.1.9's password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-31-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Sun Sep 20 09:02:08 PM UTC 2026

System load:  0.04               Processes:            222
Usage of /:   51.0% of 9.75GB    Users logged in:     0
Memory usage: 25%               IPv4 address for ens33: 192.168.1.9
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

13 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

saraajmal30@ubuntu:~$ |

```

- Demonstrates successful authentication and remote terminal login to Ubuntu Server over SSH.

```
saraajmal30@ubuntu:~$ |
```

- Displays the remote shell prompt taaniszahra51@ubuntu:~\$ confirming active SSH session identity.

➤ Basic Linux Commands

```

saraajmal30@ubuntu:~$ whoami
saraajmal30
saraajmal30@ubuntu:~$ hostname
ubuntu
saraajmal30@ubuntu:~$ date
Mon Sep 21 02:34:36 PM UTC 2026
saraajmal30@ubuntu:~$

```

EXERCISES

➤ Exercise 1: Record RAM and Disk Space

```
saraajmal30@ubuntu:~$ free -h
              total        used        free      shared  buff/cache   available
Mem:            1.6Gi         475Mi        773Mi        1.2Mi        499Mi        1.1Gi
Swap:           1.6Gi           0B         1.6Gi
saraajmal30@ubuntu:~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
tmpfs            319M  1.3M  318M   1% /run
/dev/mapper/ubun--vg-ubuntu--lv  9.8G  4.7G  4.6G  51% /
tmpfs            798M   0  798M   0% /dev/shm
none            1.0M   0   1.0M   0% /run/credentials/systemd-journald.service
none            1.0M   0   1.0M   0% /run/credentials/systemd-resolved.service
tmpfs            798M   0  798M   0% /tmp
/dev/sda2        1.8G  100M  1.6G   7% /boot
none            1.0M   0   1.0M   0% /run/credentials/systemd-networkd.service
none            1.0M   0   1.0M   0% /run/credentials/getty@tty1.service
tmpfs            160M   8.0K  160M   1% /run/user/1000
saraajmal30@ubuntu:~$
```

➤ Exercise 2: Create a Folder and File

```
saraajmal30@ubuntu:~$ mkdir mylab
saraajmal30@ubuntu:~$ cd mylab
saraajmal30@ubuntu:~/mylab$ echo Hello Cloud > hello.txt
saraajmal30@ubuntu:~/mylab$ cat hello.txt
Hello Cloud
saraajmal30@ubuntu:~/mylab$
```

➤ Exercise 3: Find IP Address

```
eth0: flags=4163<UP,BROADCAST,MULTICAST> mtu 1500
    inet 192.168.1.5/24 metric 100 brd 192.168.1.255 scope global dynamic ens33
```

➤ Exercise 4: Compare Startup Time

How long did Ubuntu Server take to start, and how long does Windows take?

The Ubuntu Server installation has **no graphical desktop environment**, so it generally has fewer graphical components and background services to initialize than a full desktop operating system.

Home Task

Explain: Difference between a virtual machine and a real computer.

Real computer: A physical machine with actual CPU, RAM, storage, motherboard, etc.

Virtual machine: A software-created computer that uses resources such as CPU, RAM, and storage provided by a physical computer.

Write five commands that are not already used in the lab.

Command	Purpose
touch	Creates an empty file
cp	Copies a file
mv	Moves or renames a file

Command	Purpose
head	Displays the beginning of a file
clear	Clears the terminal screen

Why Cloud Companies Use Linux. Give three reasons.

1. Open source

Linux is open-source, so organizations can use and customize it without paying traditional proprietary operating-system licensing fees.

2. Stability

Linux is widely used for servers because it is designed to run continuously and provides extensive server-management capabilities.

3. Resource efficiency

Linux server installations can operate without a graphical desktop, reducing unnecessary resource usage and making them suitable for server environments.

Review Questions(Viva)

Q1. What is a server?

Answer:

A server is a computer that provides services, data, or resources to other computers over a network.

Q2. What does ISO mean?

Answer:

ISO is a disk-image file that contains the files and structure of an optical disc or installation media. We use the Ubuntu ISO to install Ubuntu inside the VM.

Q3. Why doesn't Ubuntu Server have a desktop?

Answer:

Ubuntu Server normally uses a command-line interface instead of a graphical desktop because servers can operate without graphical components and can dedicate more resources to server services.

Q4. Dynamically allocated vs fixed-size disk?

Answer:

Dynamically allocated: The virtual disk file grows as data is added, up to the maximum size.

Fixed-size: The maximum disk space is allocated in advance.

For our lab we selected:

Dynamically allocated.

Q5. Why install OpenSSH?

Answer:

OpenSSH allows administrators to securely access and manage the server remotely through an SSH connection.

Q6. Why shouldn't we close the VM while Ubuntu is running?

Answer:

Because abruptly closing or powering off the VM is similar to suddenly removing power from a physical computer and can cause data loss or filesystem problems.

Q7. What does `sudo` mean?

Answer:

`sudo` allows an authorized user to execute a command with administrator/root privileges.

➤ Safely Shut Down Ubuntu

```
saraajmal30@ubuntu:~/mylab$ sudo shutdown -h now  
[sudo: authenticate] Password:
```